

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 14

Munhyeok Kang

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Topic

Automated Parameter Calibration

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Automated Parameter Calibration. State all assumptions and conventions.

$$\theta_{r+1} = \theta_r + \Delta\theta_r$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$L_R = \sum_{a=1}^{52} R_a^2$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$score = (L_R, N_{regress}, d_{SGT9})$$

Show enough intermediate work that another student can reproduce the calculation.